Execution Cost Lab
TWAP vs Forecast-Profile VWAP on Intraday Equity Bars under Participation Constraints

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Abstract

When an institutional investor executes a large buy order, the order must be split into smaller slices spread over time to avoid moving the price. The two standard scheduling policies are TWAP (uniform slicing) and VWAP (volume-weighted slicing). This paper builds a simulation framework to quantify whether a historical intraday volume profile improves execution cost relative to a naïve TWAP.

I introduce a three-way decomposition—TWAP, forecast-profile VWAP (tradeable), and realized-volume VWAP (oracle, not tradeable)—that separates the value of volume information from the cost of forecast error. Over 1,000 simulated 60-minute windows on five US equities, the forecast VWAP wins 52.5% of windows on implementation shortfall, with a mean saving of 0.0044 USD/share. A regime analysis reveals that this advantage is concentrated in high-volume environments and reverses in low-volume, low-volatility conditions. The full pipeline is implemented as a companion Python codebase.

1 Introduction

Suppose a portfolio manager decides to buy 50,000 shares of NVDA. If the full order is sent at once, it would consume all available liquidity at the current best ask price, then sweep through deeper levels of the order book, pushing the price higher with each fill. This price degradation—called *market impact* [1]—is not a theoretical concern: Berkowitz et al. [7] estimated that execution costs on the NYSE can reach 1–2% of trade value, far exceeding explicit commissions.

To mitigate impact, the order is split into smaller child orders spread over a time horizon [2]. The question becomes: how should these slices be distributed? This is the core problem of *algorithmic execution* [8].

I compare three scheduling strategies: TWAP (uniform), forecast-profile VWAP (based on a historical volume curve), and oracle VWAP (based on actual realized volume, not implementable in practice). The paper makes three contributions:

1. A clean separation between the tradeable forecast and the non-tradeable oracle, with the gap measuring the *cost of forecast error*—a diagnostic that indicates whether effort should go into better forecasting or into other aspects of execution.
2. A regime analysis showing that the VWAP advantage is not uniform across market conditions: it is strongest in high-volume environments and can disappear when both volume and volatility are low.
3. A fully reproducible pipeline (modular Python codebase, fixed seed) where every number and figure in this paper can be regenerated.

All simulations are buy-side: positive implementation shortfall means I paid more than the arrival price (unfavourable).

2 Background

2.1 The participation cap

When an algorithm executes a buy order, it must not consume too large a fraction of the available volume on any given bar. If the algorithm buys 30% of a bar's total volume, this is highly visible to other market participants and creates significant price impact. In practice, execution algorithms cap their participation at 5–20% of each bar's volume.

In this paper, I use a cap of $\rho = 5\%$: at any bar t with market volume V_t, the algorithm can execute at most $0.05 \times V_t$ shares. This is a conservative, realistic constraint. If the algorithm's desired quantity exceeds this cap, part of the order is left unexecuted on that bar.

2.2 The U-shaped intraday volume pattern

Trading volume is not uniformly distributed throughout the day. On most equity exchanges, volume follows a distinctive U-shape [4, 5]: high volume at the open (09:30–10:00, driven by overnight information and institutional order flow), low volume at midday (12:00–14:00), and high volume again at the close (15:00–16:00, driven by portfolio rebalancing and index tracking).

This pattern is the central motivation for VWAP scheduling. If the algorithm knows the volume shape in advance, it can concentrate execution during high-volume periods (where it consumes a smaller fraction of available liquidity) and reduce execution during quiet periods (where even a small order creates a large footprint).

3 Data

3.1 Universe

I use 5-minute OHLCV bars for five instruments: AAPL, MSFT, AMZN, NVDA (mega-cap tech stocks) and SPY (S&P 500 ETF), downloaded via `yfinance` over 60 trading days. This universe offers a range of liquidity profiles: SPY has extremely deep, relatively flat intraday volume; NVDA has higher volatility and a more pronounced U-shaped volume curve.

3.2 Cleaning

Raw timestamps arrive in UTC with mixed timezone offsets across DST transitions (EST –05:00 in winter, EDT –04:00 in summer). The pipeline: (1) converts to `US/Eastern` time; (2) deduplicates; (3) keeps weekdays only; (4) filters to regular trading hours 09:30–16:00 ET; (5) removes zero-volume bars. Step (1) is critical because the volume profile is defined relative to local trading hours. Step (5) prevents division-by-zero in VWAP computations.

After cleaning, the dataset contains approximately 4,500 bars per ticker, spanning ∼60 trading sessions over roughly three calendar months. Each session has 78 bars (one per 5-minute interval from 09:30 to 16:00).

4 Model

4.1 Setup

A buy order of Q shares is executed over $N = 12$ consecutive 5-minute bars (one hour). The order size is set as 3% of the total volume traded during the window:

$$Q = 0.03 \times \sum_{t=1}^{N} V_t \,. \tag{1}$$

This is a moderate order size. Since the participation cap allows up to 5% of total volume, full execution is feasible in aggregate—but uneven volume distribution across bars can prevent full fill in practice.

Remark 1 (Design choices) $\alpha = 0.03$ *represents a moderate institutional order (3% of the window's total volume).* $\rho = 0.05$ *is a conservative participation cap, typical of institutional execution algorithms.* $N = 12$ *(one hour) is a standard execution horizon for medium-sized orders. The no-catch-up rule is deliberate (see below).*

4.2 Shared execution rule

All three strategies compute a desired quantity d_t at each bar, then execute the actual quantity:

$$q_t = \min(d_t,\ \rho V_t,\ Q_{\text{rem}})\,, \tag{2}$$

where $Q_{\text{rem}} = Q - \sum_{s<t} q_s$ is the unfilled remainder. The three terms enforce, respectively: (i) the strategy's scheduling rule, (ii) the participation cap (never exceed 5% of the bar's volume), and (iii) the no-overfill constraint (stop once the full order is filled). Execution is priced at each bar's close price P_t.

Importantly, there is *no catch-up*: if the cap prevents filling bar t's target, the shortfall is not reallocated to subsequent bars. This isolates the schedule's quality from reallocation heuristics.

4.3 Strategy 1: TWAP

The desired quantity is uniform:

$$d_t^{\text{TWAP}} = Q/N\,. \tag{3}$$

TWAP is simple and requires no market information. Its weakness: it attempts to trade the same quantity during a quiet midday bar (where volume may be 50,000 shares) as during a busy opening bar (where volume may be 500,000 shares). On the quiet bar, the participation cap binds and the order is partially unfilled.

4.4 Strategy 2: Realized-volume VWAP (oracle)

The desired quantity is proportional to the bar's actual volume:

$$d_t^{\text{oracle}} = Q \cdot V_t \Big/ \sum_s V_s\,. \tag{4}$$

This means: trade a lot when the market is active, trade little when it is quiet. The participation rate q_t/V_t is constant across bars (before the cap binds), which is the theoretically ideal behaviour.

However, this schedule requires knowing V_t before bar t completes—information that is not available in real time. It is therefore *not tradeable* and serves purely as an oracle: the best any volume-proportional schedule could achieve with perfect foresight.

4.5 Strategy 3: Forecast-profile VWAP

To build a tradeable VWAP, I estimate the intraday volume shape from historical data. The idea of distributing an order proportionally to an expected volume profile follows a long line of work on optimal VWAP slicing [6].

Definition 1 (Intraday volume profile) *For each past trading session d, compute the fraction of daily volume at each bar index k: $\varphi_{d,k} = V_{d,k}/\sum_s V_{d,s}$. Average across sessions: $\bar{\varphi}_k = D^{-1}\sum_d \varphi_{d,k}$. This curve captures the U-shape.*

For a window starting at bar index k_0, the desired quantity is:

$$d_t^{\text{fcst}} = Q \cdot \frac{\bar{\varphi}_{k_0+t-1}}{\sum_s \bar{\varphi}_{k_0+s-1}}\,. \tag{5}$$

In words: bars whose historical profile predicts higher volume receive a proportionally larger slice of the order. The profile is built from sessions strictly *before* the window's date, ensuring the forecast is genuinely ex-ante.

4.6 Metrics

Implementation shortfall (IS), introduced by Perold [3], is the standard metric for execution quality. It measures the difference between the average price actually paid and the price at which the decision was made (the arrival price, here defined as the first bar's close):

$$\mathrm{IS} = \bar{P}_{\mathrm{exec}} - P_0, \quad \bar{P}_{\mathrm{exec}} = \frac{\sum_t q_t P_t}{\sum_t q_t}. \tag{6}$$

For a buy order, IS > 0 means the market moved up during execution—I paid more than at decision time (unfavourable). IS < 0 means the market moved down during execution (favourable). IS captures all implicit costs—impact, timing, drift—in a single number expressed in USD/share. It is computed on executed shares only.

Fill ratio: Fill $= \sum_t q_t/Q$. Values below 1 indicate that the participation cap prevented full execution. Fill is tracked separately from IS to avoid conflating schedule quality with underfill: a strategy that fills only 80% of the order at a good price should not be confused with one that fills 100% at a bad price.

5 Experimental Design

5.1 Pipeline structure

The codebase consists of six Python modules executed sequentially: `download_data` (fetch OHLCV from Yahoo Finance), `clean_data` (timezone conversion and filtering), `execution` (TWAP/VWAP simulators and metric computation), `run_experiment` (window sampling and strategy comparison), `make_plots` (summary tables and figures), and `run_one_window` (single-window visual demo). A shell script `run_all.sh` orchestrates the full pipeline end-to-end.

5.2 Window sampling

For each ticker, I enumerate all valid contiguous 12-bar windows within each trading session and sample 200 at random (`seed` = 42), yielding 1,000 windows across five tickers. All bars in a window come from a single session—mixing sessions would break the arrival price reference and distort the volume profile.

Each window is characterised by two conditioning variables used in the regime analysis: realized volatility $\sigma = \sqrt{\sum r_t^2}$ (from 5-minute log-returns) and volume CV $= \sigma_V/\mu_V$ (how unevenly volume is distributed across the 12 bars). High CV means some bars are much more active than others—exactly the condition where VWAP should outperform TWAP.

6 Results

6.1 Single-window illustration

Figure 1 shows one 60-minute window on AAPL, covering the opening session from 09:30 to 10:25. The top panel shows the market volume per bar with the 5% participation cap as a dotted line—note the massive opening-bar volume, typical of the intraday U-shape. The second panel compares the per-bar execution schedules. TWAP targets a constant $Q/N \approx 417$ shares per bar. The forecast VWAP, using a stylised U-shaped profile for illustration, leans slightly toward the first and last bars. The oracle VWAP, which sees actual volume, concentrates a large fraction of the order on the first bar, where volume is highest. In this particular window, $Q = 5{,}000$ is small relative to market volume, so the participation cap never binds on any strategy—the panel therefore illustrates the *shape differences* between schedules rather than cap-induced underfill. The cap-binding phenomenon is real in aggregate (see the fill ratios below 100% in Table 1), but it does not materialise on this particular window. The third panel shows cumulative fill; the bottom panel shows the price path.

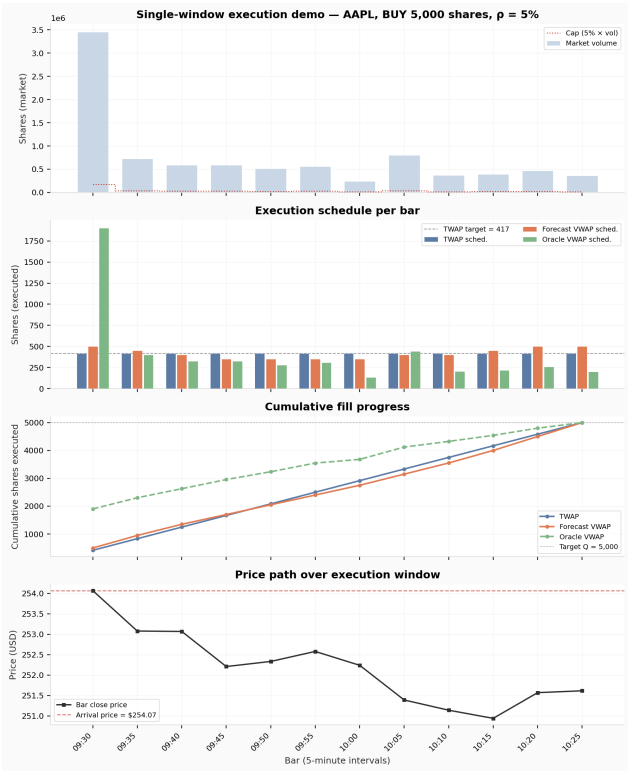

Figure 1: Single-window demo (AAPL). Top: market volume per bar with the 5% participation cap. Second: per-bar execution schedule for each strategy. Third: cumulative fill. Bottom: price path.

6.2 Aggregate results

Table 1 reports per-ticker averages over the 1,000 windows. Win rates range from 48.5% (AAPL) to 55.0% (SPY). The advantage is real but modest.

A note on negative IS values. Several tickers show negative mean IS (e.g., AAPL: -0.047, SPY: -0.067). This does *not* mean the strategies are generating profit—it means the market happened to move down on average during the sampled windows while the buy order was executing, a favourable coincidence driven by the sampled price paths. The meaningful comparison is always

Table 1: Per-ticker summary (IS in USD/share).

	TWAP IS	Fcst IS	TWAP fill	Fcst fill	Win %
AAPL	−.047	−.045	.904	.899	48.5
AMZN	+.005	−.001	.947	.933	53.5
MSFT	+.012	−.003	.937	.930	53.0
NVDA	−.018	−.020	.950	.934	52.5
SPY	−.067	−.069	.986	.988	55.0

the difference ΔIS between strategies, which cancels out the common price drift.

Three observations:

SPY has the highest win rate. SPY is the most liquid instrument in the universe, with a very stable intraday volume profile. When the shape is strong and repeats across sessions, the historical average captures it reliably, so the forecast beats TWAP on more windows than any other ticker—though the average magnitude of the improvement is modest.

AAPL is below 50%. AAPL's win rate (48.5%) is the only one under the baseline. Two factors likely contribute. First, AAPL has the lowest fill ratio (90.4%): the participation cap binds often, which penalises the forecast VWAP more than TWAP because concentrating execution makes the schedule more exposed when the bet is wrong. Second, Figure 4 shows that on AAPL during this sample period even the oracle—which has perfect volume information—produces a slightly *worse* IS than TWAP, suggesting that high-volume bars tended to coincide with higher prices (adverse for a buy order). Both factors push in the same direction: volume-proportional scheduling was not beneficial on AAPL in this dataset.

Fill ratios are slightly lower for the forecast VWAP on all tickers except SPY. This trade-off is discussed in Section 7.2.

6.3 The IS difference distribution

Figure 2 shows $\Delta\text{IS} = \text{IS}_{\text{fcst}} - \text{IS}_{\text{TWAP}}$ per window. The distribution is centred slightly left of zero with mean ≈ -0.0044 USD/share, and the forecast VWAP wins 52.5% of windows. The wide spread—individual windows reach roughly ±0.5 USD/share in the tails—reveals an important insight: on any single window, whether the market moved up or down during the hour matters far more than the scheduling choice. The scheduling effect is a second-order improvement on top of first-order price randomness.

6.4 Regime analysis: when does VWAP help?

This is the central empirical finding. I split the 1,000 windows into four quadrants by the median of window volume (V_{60}) and realized volatility (σ). Table 2 reports the results.

The pattern is clear. In **high-volume** environments (top two rows), the forecast VWAP outperforms TWAP consistently (56.6% and 54.6% win rates): the intraday

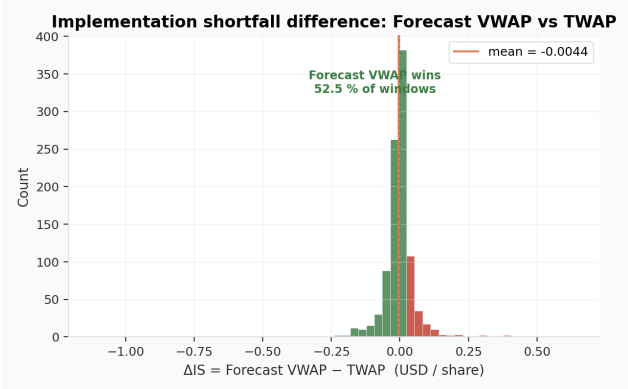

Figure 2: ΔIS distribution over 1,000 windows. Green = forecast wins.

Table 2: Regime analysis.

Volume	Volatility	n	$\overline{\Delta\text{IS}}$	Win%
High	High	251	−.0092	56.6
High	Low	249	−.0046	54.6
Low	High	249	−.0046	50.6
Low	Low	251	+.0009	48.2

pattern is strong, and the historical profile captures it well enough to load execution into liquid bars. In the **low-volume, low-volatility** quadrant, TWAP is marginally better (48.2% win rate, $\Delta\text{IS} = +0.0009$): when both the market and the price path are calm, there is little U-shape to exploit, and concentrating execution based on a flat-ish forecast is slightly worse than spreading it uniformly.

The key driver is volume, not volatility: moving from low to high volume lifts the win rate by about 6 percentage points at each volatility level (48.2% → 54.6% and 50.6% → 56.6%), while moving from low to high volatility lifts it by only 2–2.4 points. This matches the intuition that VWAP's mechanism of action is *aligning with liquidity*, and liquidity is what high volume provides.

Figure 3 visualises this as a heatmap.

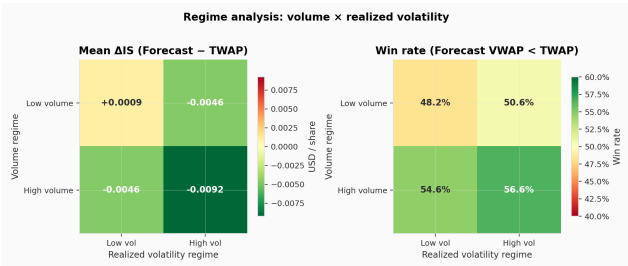

Figure 3: Regime heatmap. Left: mean ΔIS. Right: win rate. High-volume rows favour the forecast; low-volume rows are closer to the baseline.

6.5 Oracle decomposition

The three-way comparison (Figure 4) enables a diagnostic decomposition that is the paper's key conceptual contribution:

- **Oracle vs TWAP:** the maximum possible benefit

of knowing the volume shape perfectly. This is an upper bound—no real strategy can exceed it.

- **Forecast vs TWAP:** the actual benefit delivered by the historical average profile.
- **Forecast vs Oracle:** the *cost of forecast error*. If this gap is large, a better forecast (exponential weighting, day-of-week conditioning) would help. If it is small, the forecast is already close to the best possible.

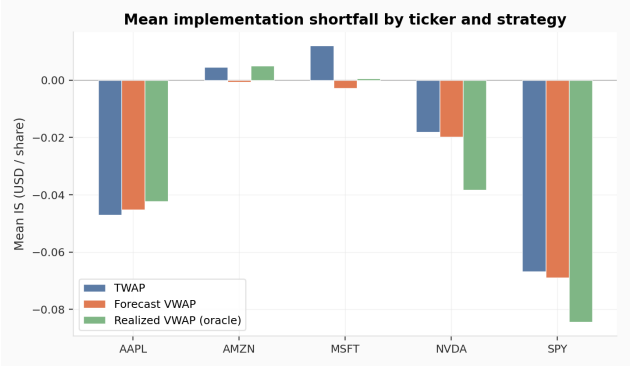

Figure 4: Mean IS by strategy. The forecast-to-oracle gap is the cost of forecast error.

The picture varies substantially by ticker. On **NVDA and SPY**, the oracle substantially beats TWAP (by about 0.020 and 0.017 USD/share); the forecast captures only a small fraction of this gap, reflecting that the historical average approximates but does not equal the true daily volume shape. On **AMZN and MSFT**, the forecast VWAP is actually the best of the three—even beating the oracle—suggesting that a smooth historical profile can dominate the realized-day shape when the realized shape carries adverse price–volume co-movement. On **AAPL**, the oracle is marginally *worse* than TWAP: during this sample period, high-volume bars coincided with higher prices, making concentration costly. This is a useful reminder that oracle-VWAP is optimal for aligning *participation* with liquidity, not for minimising *cost*.

This decomposition is actionable. When the oracle-to-TWAP gap is large and positive (NVDA, SPY), scheduling is a meaningful lever and investing in a better volume forecast would pay off. When the gap is near zero or negative (AAPL), scheduling is not the bottleneck—other execution costs (spread, impact) deserve priority.

6.6 Supporting figures

Figure 5 shows IS distributions. The within-strategy spread (interquartile range) is much larger than the between-strategy gap, confirming that market conditions dominate scheduling effects.

Figure 6 shows fill ratios. The forecast VWAP fills slightly less than TWAP on all tickers except SPY, reflecting the cap-violation trade-off discussed below.

Figure 7 shows win rates by ticker with the 50% baseline. No ticker is dramatically above or below, confirming that the effect is systematic but small.

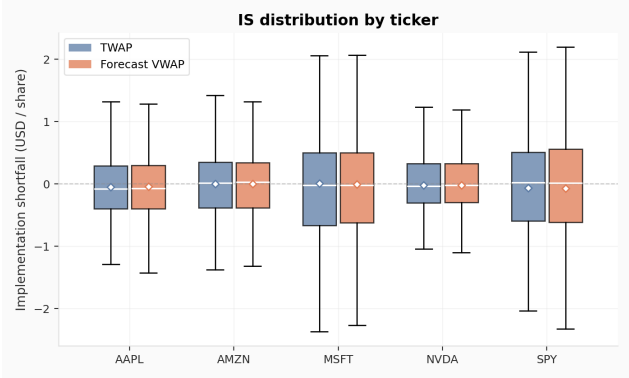

Figure 5: IS distributions. The IQR dwarfs the strategy gap.

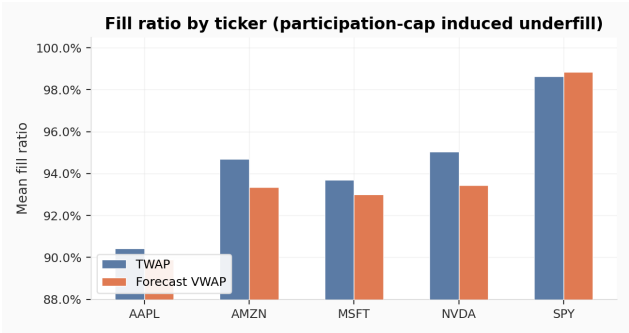

Figure 6: Fill ratios by ticker.

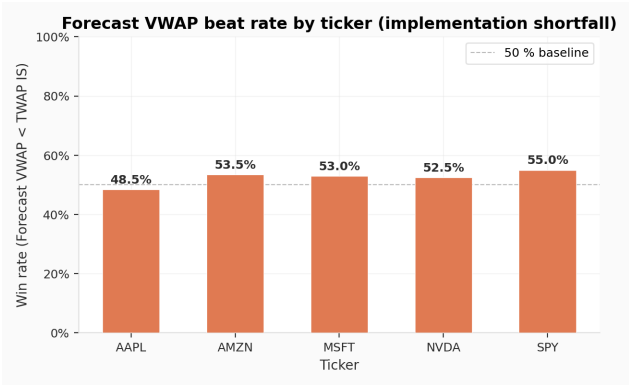

Figure 7: Win rate by ticker (dashed = 50% baseline).

7 Discussion

7.1 Why the effect is small

A 52.5% win rate and a 0.0044 USD/share mean advantage may seem disappointing. This is expected. In this simulation, execution is priced at bar close and there is no explicit market impact model. The only way VWAP can outperform is by interacting with the price–volume correlation within each window. This effect is inherently noisy because 60-minute price paths are dominated by random walk behaviour at the 5-minute scale.

An explicit impact model—even a simple one where execution cost is proportional to participation rate—would amplify the VWAP advantage significantly. Under such a model, TWAP's habit of pushing into low-volume bars (where it consumes a large fraction of available liquidity) would be directly penalised, while

VWAP's alignment with volume would be rewarded. This is the most impactful extension of the current framework.

7.2 The fill ratio trade-off

The forecast VWAP's slightly lower fill ratio (e.g., 93.4% vs 95.0% on NVDA) is not a flaw—it is a structural consequence of the strategy. By concentrating execution, VWAP is "betting" on the forecast being correct. When it is right (high-volume regimes), the concentration pays off. When it is wrong (low-volume bars that the forecast expected to be busy), the cap binds and the order is underfilled. TWAP, by spreading uniformly, makes no forecast bet and is therefore more robust to forecast error—at the cost of never exploiting the volume shape.

7.3 Practical implications

The regime analysis points toward an **adaptive strategy**: use VWAP during periods when volume is expected to be high (first and last hours of the trading day, earnings weeks) and switch to TWAP when both volume and volatility are low (calm midday sessions on quiet tickers). An algorithm that monitors real-time volume CV and switches scheduling rule accordingly could capture the VWAP advantage when it exists while avoiding the conditions where it underperforms.

8 Limitations and Extensions

Limitations.

1. All trades priced at bar close—no intra-bar price dynamics, limit-order simulation, or bid-ask spread.
2. No market impact model; the participation cap is the sole proxy.
3. No catch-up for underfilled bars—a real algorithm would redistribute remaining shares.
4. IS is computed on executed shares only, which can mask underfill.
5. The volume forecast is a plain historical average, with no exponential decay, rolling window, or day-of-week conditioning.
6. No explicit transaction costs (commissions, exchange fees).

Extensions.

1. Add a temporary impact term proportional to participation rate and study how the optimal schedule changes.
2. Replace the historical average with an exponentially-weighted forecast and measure the reduction in the forecast-to-oracle gap.
3. Add a Percentage-of-Volume (POV) strategy that targets a constant participation rate as a third benchmark.
4. Build an adaptive TWAP/VWAP switcher conditioned on real-time volume CV.
5. Test with sell orders, larger order sizes, and varying participation caps.
6. Apply to futures, FX, or less liquid equities where volume patterns may differ.

9 Conclusion

I presented a reproducible comparison of three execution schedules—TWAP, forecast-profile VWAP, and oracle VWAP—on intraday equity bars. The forecast VWAP achieves a modest but systematic advantage over TWAP (win rate 52.5%, mean saving 0.0044 USD/share over 1,000 windows), concentrated in high-volume regimes where the intraday volume pattern is strong and predictable. The advantage shrinks or reverses when both volume and volatility are low, conditions under which the profile contains little exploitable structure.

The oracle decomposition (Section 6.5) provides an actionable diagnostic per ticker. In this dataset, both the value of perfect volume information and the forecast's capture of that value are small, suggesting that the highest-return next step is not refining the volume forecast but incorporating explicit market impact into the simulation.

The full pipeline is available as an open-source Python codebase. Every number and figure in this paper can be reproduced with `bash run_all.sh`.

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